

Advanced Concepts in Large Language Models

DATA 443/643

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Classroom:	Date/Time:
Office Hours:	Office: DMTI - 106J

1 Overview

1.1 Course description:

DATA 443/643– *Advanced Concepts in Large Language Models* is an upper-level course exploring the theory, architecture, and practice of modern artificial intelligence (AI) systems for language understanding and generation. The course situates large language models within the broader AI ecosystem—representation learning, multimodal perception, alignment, reasoning, and responsible deployment. Students gain both mathematical and practical understanding of how AI-based language models are designed, trained, and adapted. Labs and a final project emphasize research applications in instruction tuning, retrieval-augmented generation, continual learning, and multimodal AI integration.

Pre-requisite: DATA 427 “Statistical Machine Learning” or DATA 627 “Statistical Machine Learning”.

1.2 Learning outcomes (LO) and assessments (AS):

1. LO: Explain and implement the key architectures underlying modern AI and language models (Transformers, GPT, BERT, multimodal systems).
 - AS: Homework assignments and midterm exam
2. LO: Apply optimization and training strategies to fine-tune or align AI-driven language systems.
 - AS: Homework and in-class labs.
3. LO: Critically evaluate research papers and findings in multimodal AI and model alignment.
 - AS: Quizzes and discussions.
4. LO: Build and test efficient, safe, and adaptive AI language systems using open-source frameworks.
 - AS: In-class Labs and final project.
5. LO: Present and defend AI research ideas clearly and effectively.
 - AS: Project proposal, oral presentation, and final report.

Graduate students (DATA-643) will, in addition, be able to:

6. (DATA 643 Only) LO: Independently investigate and implement ideas from a recent research paper (2024–2025) related to large language models or multimodal AI.
 - AS: Select one peer-reviewed or preprint paper, reproduce or extend one of its experiments using open-source resources, and submit a 3-page report plus code notebook.
7. (DATA 643 Only) LO: Critically evaluate the chosen research paper's methods, results, and limitations in the context of AI model design and alignment.
 - AS: Oral presentation (≈ 10 minutes) summarizing findings and leading a short Q&A during the semester.

1.3 Materials:

Core readings span contemporary artificial intelligence and machine learning literature, emphasizing architecture, training, and interpretability of large-scale models.

1. Transformers for Natural Language Processing , Hugging Face Team, O'Reilly Media, 2022.
2. Deep Learning , Goodfellow, Bengio, Courville, MIT Press.
3. Hands-On Artificial Intelligence with Transformers and Diffusion Models , Packt Publishing, 2024.
4. Interpretable Machine Learning , Molnar (online edition 2025).
5. Reinforcement Learning from Human Feedback , Lambert (2024 Edition).
6. Additional readings and papers will be provided on Canvas.

1.4 Software:

Python 3, PyTorch, Hugging Face Transformers, Jupyter Notebooks—core tools for modern AI research.

2 Modules and Weekly Topics:

2.1 Module 1: Foundations of Artificial Intelligence (Weeks 1–3)

2.1.1 Week 1: Neural Networks & Word Embeddings

- Topics: Neurons, activations, feedforward networks, gradient descent, distributed representations, semantic geometry in AI.
- Discussion: Intrinsic vs. extrinsic evaluation of embeddings; bias and interpretation in AI models.

2.1.2 Week 2: Sequence Models — RNNs, LSTMs, and GRUs

- Topics: Sequential modeling, vanishing/exploding gradients, gating mechanisms, hidden state dynamics.
- Discussion: Temporal dependencies and limitations of recurrent models; transition to attention in modern AI.

2.1.3 Week 3: Attention Mechanisms & Transformers

- Topics: Self-attention (Q/K/V), positional encoding, encoder–decoder design, context modeling.
- Discussion: Architecture and pretraining objectives of GPT and BERT as AI systems; scalability and efficiency.

2.2 Module 2: Multimodal AI Systems (Weeks 4–6)

2.2.1 Week 4: Multimodal Foundations & Vision Transformers —

- Topics: Patching images as tokens; contrastive learning; modality alignment; zero-shot transfer in multimodal AI.

2.2.2 Week 5: Vision–Language Models (CLIP & BLIP) —

- Topics: Cross-modal retrieval; captioning; bootstrapped pretraining; bias and robustness.

2.2.3 Week 6: Audio & Cross-Modal Integration —

- Topics: Audio tokenization as spectrograms; multimodal fusion; unified embeddings; data scarcity in AI systems.

2.3 Module 3: Core AI Architecture, Alignment & Adaptation (Weeks 7–10b)

2.3.1 Week 7: Scaling, Efficiency & Deployment —

- Topics: Scaling laws; mixture-of-experts; pruning/quantization/distillation; speculative decoding for efficient AI systems.

2.3.2 Week 8a: Alignment I — Supervised Fine-Tuning (SFT) —

- Topics: Instruction tuning; dataset design; prompt engineering; task generalization in AI.

2.3.3 Week 8b: Continual & Federated Learning —

- Topics: Continual updates; catastrophic forgetting; adapter-based fine-tuning (LoRA); federated optimization.

2.3.4 Week 9: Alignment II — RLHF & Direct Preference Optimization —

- Topics: Reward modeling; policy optimization; preference learning; human feedback in AI alignment.

2.3.5 Week 10a: Safety, Control & Knowledge Editing —

- Topics: Safety fine-tuning; red-teaming; machine unlearning; knowledge editing (ROME, MEMIT, MEND).

2.3.6 Week 10b: Retrieval-Augmented Fine-Tuning (RAFT) & Dynamic Memory —

- Topics: Integrating retrieval into training; memory-augmented models (RETRO, MemGPT); long-term consistency in AI models.

2.4 Module 4: Generative AI & Advanced Capabilities (Weeks 11–14)

2.4.1 Week 11: Generative Models I — GANs —

- Topics: Adversarial training; generator–discriminator dynamics; mode collapse in generative AI.

2.4.2 Week 12: Generative Models II — Diffusion Models —

- Topics: Forward/reverse diffusion; latent diffusion; text-to-image conditioning; sampling trade-offs in AI generation.

2.4.3 Week 13: Reasoning & Chain-of-Thought —

- Topics: Step-by-step reasoning; self-consistency; Tree-of-Thoughts; logical faithfulness in AI reasoning.

2.4.4 Week 14: Tool Use, Retrieval-Augmented Generation & Final Presentations —

- Topics: RAG pipelines; ReAct framework; uncertainty calibration; evaluation of AI-enabled applications.

3 Assessment

3.1 Assignments & Grading:

- Assignments (25%) — Homework per module including mathematical and coding tasks. Late submissions incur 10% per day before solutions release.
- Quizzes (10%) — Short reading quizzes at module ends.
- Midterm Exam (10%) — Conceptual and computational.
- Labs (10%) — Weekly programming labs in Python or PyTorch.
- (Undergraduate) Final Project (30%) = Proposal (10%) + Presentation (10%) + Report (10%) — Group research project in artificial intelligence or large language models, culminating in a presentation and 10-page IEEE-format report.
- (Graduate) Final Project (20%) = Proposal (5%) + Presentation (10%) + Report (5%) — Independent research project in artificial intelligence or large language models, culminating in a presentation and 10-page IEEE-format report.
- (Graduate) Mini-Final Project (10%) = Proposal (2.5%) + Presentation (5%) + Report (2.5%) — Individual mini-project based on a recent (last 6 months) research paper; reproduce or extend one of its experiments and present findings in a 3-page IEEE-format report.
- Final Exam (10%) = Cumulative Exam
- Participation & Engagement (5%) = Attendance and discussion.

3.2 Grading Policy:

93–100% = A, 90–92% = A–, 88–89% = B+, 83–87% = B, 80–82% = B–, 78–79% = C+, 70–77% = C, 60–69% = D, below 60% = F.

3.3 Important Dates (Fall 2026):

Midterm	October –, 2026
Project Proposal	October –, 2026
(Graduate) Mini-Project Proposal	October –, 2026
(Final & Mini-) Project Presentations	December –, 2026
Report	December –, 2026
Final Exam	December –, 2026

4 Additional Notes:

1. Be courteous during lectures and discussions.
2. Notify the instructor early if you require accommodations.
3. Attendance and participation are expected.
4. Independent research (e.g., Hugging Face docs, StackOverflow) is part of learning in this AI-related course. This course prepares students to contribute to cutting-edge research and responsible innovation in artificial intelligence.

Academic Dishonesty:

Academic dishonesty is a serious offense. You may collaborate with other students, but your submissions must be your own. Cite all sources and credit collaborators. If you cannot explain your work in your own words, no credit will be awarded. Refer to AU's Integrity Code for details: <https://www.american.edu/academics/integrity/code.cfm>.

5 University-Wide Policies

5.1 Academic Integrity

Standards of academic conduct are set forth in the university's Academic Integrity Code. By registering for this course, students have acknowledged their awareness of the Academic Integrity Code and they are obliged to become familiar with their rights and responsibilities as defined by the Code. Violations of the Academic Integrity Code will not be treated lightly and disciplinary action will be taken should violations occur. This includes cheating, fabrication, and plagiarism.

5.2 Defining and Reporting Discrimination and Non-Title IX Sexual Misconduct Policy

American University is an equal opportunity, affirmative action institution that operates in compliance with applicable laws and regulations. The University does not discriminate on the basis of race, color, national origin, religion, sex, pregnancy or parenting, age, sexual orientation, disability, marital status, personal appearance, gender identity and expression, family responsibilities, political affiliation, source of income, veteran status, an individual's genetic information or any other bases under applicable federal and local laws and regulations (collectively "Protected Bases") in its programs and activities. The University expressly prohibits any form of discriminatory harassment including sexual harassment, dating and domestic violence, rape, sexual assault, sexual exploitation and stalking.

As a faculty member, I am required to report discriminatory or harassing conduct to the university if I witness it or become aware of it – regardless of the location of the incident. There are four confidential resource on campus if you wish to speak to someone who is not required to report: Counseling Center, victim advocates in OASIS, medical providers in the Student Health Center, and ordained clergy in the Kay Spiritual Life Center. If you experience any of the above, you have the option of filing a report with [University Police](#) (202-885-2527), the [Office of the Dean of Students](#) (dos@american.edu or 202-885-3300), or the [Title IX Office](#) (202-885-8080 or TitleIX@american.edu). For more information, including a list of supportive resources on and off-campus, contact [OASIS](#) (oasis@american.edu or 202-885-7070) or check out the [Support Guide on the Title IX webpage](#).

5.3 Defining and Reporting Title IX Sexual Harassment Policy

Review the complete [Discrimination and Non-Title IX Sexual Misconduct Policy](#). American University does not discriminate on the basis of sex in the education program or activity that it operates, which includes admission and employment. Consistent with the procedures set forth and referenced in this Policy, the University will take steps to eliminate Title IX Sexual 2 Harassment, prevent its recurrence, and remedy any discriminatory effects for members of the AU Community.

It is the responsibility of every member of the AU Community to foster an environment free of Title IX Sexual Harassment. All members of the AU Community are encouraged to take reasonable and prudent actions to prevent or stop an act of Title IX Sexual Harassment. Taking action may include direct intervention when safe to do so, enlisting the assistance of friends, contacting law enforcement, or seeking assistance from a person in authority.

AU Community members employed by the University (including student employees, such as teaching assistants, resident assistants, and orientation leaders) and individuals covered by the University's Out-of-State Staffing Policy have a duty to report Title IX Sexual Harassment to the [Title IX Coordinator](#) (202-885-8080), when they become aware of such conduct.

Review the complete [Title IX Sexual Harassment Policy](#).

5.4 Emergency Preparedness

In the event of an emergency, American University will implement a plan for meeting the needs of all members of the university community. Should the University be required to close for a period of time, we are committed to ensuring that all aspects of our educational programs will be delivered to our students. These may include altering and extending the duration of the traditional term schedule to complete essential instruction in the traditional format and/or the use of distance instructional methods. Specific strategies will vary from class to class, depending on the format of the course and the timing of the emergency.

Faculty will communicate class-specific information to students via AU email and Blackboard, while students must inform their faculty immediately of any emergency-related absence. Students are responsible for checking their AU email regularly and keeping themselves informed of emergencies. In the event of an emergency, students should refer to the AU Student Portal, the [AU website](#), and the AU information line at (202) 885-1100 for general university-wide information, as well as contact their faculty and/or respective dean's office for course and school/college specific information.

5.5 Incomplete Policy

At the discretion of the faculty member and before the end of the semester, the grade of I (Incomplete) may be given to a student who, because of extenuating circumstances, is unable to complete the course during the semester. The grade of Incomplete may be given only if the student is receiving a passing grade for the coursework completed. Students on academic probation may not receive an Incomplete. The instructor must provide in writing to the student the conditions, which are described below, for satisfying the Incomplete and must enter those same conditions when posting the grades for the course. The student is responsible for verifying that the conditions were entered correctly.

Conditions for satisfying the Incomplete must include what work needs to be completed, by when the work must be completed, and what the course grade will be if the student fails to complete that work. At the latest, any outstanding coursework must be completed before the end of the following semester, absent an agreement to the contrary. Instructors will submit the grade of I and the aforementioned conditions to the Office of the University Registrar when submitting all other final grades for the course. If the student does not meet the conditions, the Office of the University Registrar will assign the default grade automatically.

The Associate Dean of the Academic Unit, with the concurrence of the instructor, may grant an extension beyond the agreed deadline, but only in extraordinary circumstances. Incomplete courses may not be retroactively dropped. An Incomplete may not stand as a permanent grade and must be resolved before a degree can be awarded.

[More information on AU Regulations and Policies.](#)

5.6 Student Code of Conduct

The central commitment of American University is the development of thoughtful, responsible human beings in the context of a challenging yet supportive academic community. The [Student Code of Conduct](#) is designed to benefit the American University community and to assist in forming the highest standards of ethics and morals among its members. By registering for this course, students have acknowledged their awareness of the Student Code of Conduct and they are obliged to become familiar with their rights and responsibilities as defined by the Code.

5.7 Religious Observances

Students will be provided the opportunity to make up any examination, study, or work requirements that may be missed due to a religious observance, provided they notify their instructors before the end of the second week of classes. Please send this notification through email to the professor. For additional information, see American University's [religious observances policy](#).

5.8 Use of Student Work

The professor will use academic work that you complete for educational purposes in this course during this semester. Your registration and continued enrollment constitute your consent.

6 Academic Support Services

6.1 Academic Support

All students may take advantage of the [Academic Support and Access Center \(ASAC\)](#) for individual academic skills counseling, workshops, Tutoring and Writing Lab appointments, peer tutor referrals, and Supplemental Instruction. The ASAC is located in Mary Graydon Center 243.

Additional academic support resources available at AU include the Bender Library, the Department of Literature's Writing Center (located in the Library), the Math Lab in the Department of Mathematics & Statistics, and the Center for Language Exploration, Acquisition, & Research (CLEAR) in Anderson Hall, Room B-101. A more complete list of campus-wide resources is available in the ASAC.

6.2 International Student & Scholar Services

[International Student & Scholar Services](#) has resources to support academic success and participation in campus life including academic counseling, support for second language learners, response to questions about visas, immigration status and employment and intercultural programs, clubs and other campus resources. (202-885-3350, Butler Pavilion 410).

6.3 Writing Center

The [Writing Center](#) offers free, individual coaching sessions to all AU students. In your 45-minute session, a student writing consultant can help you address your assignments, understand the conventions of academic writing, and learn how to revise and edit your own work. (202-885-2991, Bender Library – 1st Floor Commons).

7 Student Support Services

7.1 Center for Diversity & Inclusion (CDI)

[CDI](#) is dedicated to enhancing LGBTQ, multicultural, first-generation, and women's experiences on campus and to advancing AU's commitment to respecting and valuing diversity by serving as a resource and liaison to students, staff, and faculty on issues of equity through education, outreach, and advocacy. It is located on the 2nd floor of Mary Graydon Center (202-885-3651, MGC 201 & 202).

7.2 Counseling Center

The [Counseling Center](#) offers counseling and consultations regarding personal concerns, self-help information, and connections to off-campus mental health resources. (202-885-3500, MGC 214).

7.3 Dean of Students Office

The [Dean of Students Office](#) offers individual meetings to discuss issues that impact the student experience, including academic, social, and personal matters; making referrals to appropriate campus resources for resolution. Additionally, while academic regulations state that medical absences are to be excused, if faculty require documentation to verify the student's explanation, such documentation should be submitted to the Dean of Students. The office will then receive the documentation and verify the medical excuse. Faculty have the discretion to approve absences and do not need permission from the Dean of Students to excuse absences. Students should be sent to the Dean of Students only if faculty require further proof or if they have concerns about the impact of absences on the student's ability to succeed (202-885-3300, Butler Pavilion 408).

7.4 Food and Housing Insecurity Statement

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the [Dean of Students](#) (dos@american.edu) for support. Furthermore, please notify the professor if you are comfortable in doing so. This will enable them to provide any resources that they may possess.

7.5 Office of Advocacy Services for Interpersonal and Sexual Violence

[OASIS](#) provides free and confidential advocacy services for students who have experienced sexual assault, dating or domestic violence, sexual harassment, and/or stalking. Please

email or call to schedule an appointment with a victim advocate in OASIS. (oasis@american.edu, 202885-7070, Health Promotion and Advocacy Center – Hughes Hall 105). Students can also book an appointment with one of our two confidential victim advocates.

7.6 Respect for Diversity

As stated in the [American University Discrimination and Sexual Harassment Policy](#):

"American University is an equal opportunity, affirmative action institution that operates in compliance with applicable laws and regulations. The University does not discriminate on the basis of race, color, national origin, religion, sex, pregnancy or parenting, age, sexual orientation, disability, marital status, personal appearance, gender identity and expression, family responsibilities, political affiliation, source of income, veteran status, an individual's genetic information or any other bases under applicable federal and local laws and regulations (collectively "Protected Bases") in its programs and activities. The University expressly prohibits any form of discriminatory harassment including sexual harassment, dating and domestic violence, rape, sexual assault, sexual exploitation and stalking."

The above website includes further details, including how to report instances of discrimination and your responsibilities as a member of the campus community in relation to the policy; you are strongly encouraged to familiarize yourself further with this policy.

Class rosters and University data systems are provided to faculty with the student's legal name and legal gender marker. As a student, you are able to change how your preferred/proper name shows up through email, Blackboard, and on your AU ID Card. This option is helpful for various student populations, including but not limited to: students who abbreviate their first name; students who use their middle name; international students; and transgender students. As a faculty member, I am committed to using your proper name and pronouns. We will take time during our first class together to do introductions, at which point you can share with all members of our learning community what name and pronouns you use, as you are comfortable. Additionally, if these change at any point during the semester, please let me know and we can develop a plan to share this information with others in a way that is safe for you. Should you want to update your preferred/proper name, you can do so by looking at the [guidelines and frequently asked questions](#) from the Center for Diversity and Inclusion.

7.7 Students with Disabilities

If you wish to receive accommodations for a disability, please notify me with a letter from the Academic Support and Access Center. As accommodations are not retroactive, timely notification at the beginning of the semester, if possible, is strongly recommended. To register with a disability or for questions about disability accommodations, contact the Academic Support and Access Center at 202-885-3360 or asac@american.edu, or drop by MGC 243. For more information, visit AU's [Disability Accommodations web page](#).